

Owned Go Bricks — Test Coverage Sweep (Stream BT)

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Scope

Read-only build/vet/gofmt/test sweep of the OWNED Go submodule bricks under submodules/ in the Helix OTA repo (§11.4.169 comprehensive test-type coverage evidence + §11.4.28 equal-codebase attention to owned submodules).

No source was modified. No git command was run. submodules/helixqa and submodules/http3 were SKIPPED per instruction (foreign/in-flight work owned by another stream, per §11.4.84 working-tree quiescence).

Discovery: find submodules -maxdepth 3 -name go.mod -not -path '*/vendor/*' found 14 go.mod files; excluding helixqa and http3 leaves 13 owned Go module roots swept below (note submodules/llms_verifier has a nested second module submodules/llms_verifier/llm-verifier — both are swept as distinct module roots).

Results

Module	Build	Vet	gofmt	Tests	Evidence log
submodules/ challenges	OK	OK	CLEAN	PASS (20 pkgs)	submodules_challen
submodules/ containers	OK	OK	CLEAN	FAIL (1 test, non-reproducible — see finding)	submodules_contain
submodules/ doc_processor	OK	OK	CLEAN	PASS (10 pkgs)	submodules_doc_pro
	OK	OK		PASS (9 pkgs)	submodules_llm_or

Module	Build	Vet	gofmt	Tests	Evidence log
submodules/ llm_orchestrator			NEEDS_FORMAT (6 files)		
submodules/ llm_provider	OK	OK	CLEAN	PASS (53 pkgs)	submodules_llm_pro
submodules/ llms_verifier	OK	OK	CLEAN	PASS (15 pkgs, 1 pkg “no tests to run”)	submodules_llms_ve
submodules/ llms_verifier/ llm-verifier	OK	OK	CLEAN	PASS (60 pkgs)	submodules_llms_ve verifier.log
submodules/ota- artifact- validator	OK	OK	CLEAN	PASS (3 pkgs)	submodules_ota-ar validator.log
submodules/ota- protocol	OK	OK	CLEAN	PASS (3 pkgs)	submodules_ota-pro
submodules/ota- rollout-engine	OK	OK	CLEAN	PASS (3 pkgs)	submodules_ota-ro engine.log
submodules/ota- telemetry-schema	OK	OK	CLEAN	PASS (3 pkgs)	submodules_ota-te schema.log
submodules/ security	OK	OK	CLEAN	PASS (18 pkgs)	submodules_securit
submodules/ vision_engine	OK	OK	NEEDS_FORMAT (16 files)	PASS (8 pkgs)	submodules_vision_

Module count swept: 13. All 13 build clean (go build ./... exit 0), all 13 vet clean (go vet ./... exit 0). 11/13 gofmt-clean; 2/13 need formatting. 12/13 fully green on tests; 1/13 (containers) surfaced one real test FAILURE during the full-sweep run.

Findings requiring surfacing (anti-bluff, §11.4.6 / §11.4.1)

1. submodules/containers — TestIdleShutdown_TouchResets FAILED once (real finding, non-reproducible in isolation)

During the full go test ./... -count=1 sweep of the containers module, pkg/lifecycle FAILED:

```
--- FAIL: TestIdleShutdown_TouchResets (0.37s)
    idle_test.go:37:
        Error Trace: .../pkg/lifecycle/idle_test.go:37
        Error:      Not equal:
                     expected: 0
                     actual  : 1
        Test:       TestIdleShutdown_TouchResets
FAIL
FAIL    digital.vasic.containers/pkg/lifecycle  2.184s
```

Reproducibility check performed (read-only, no source changes): re-ran the single test in isolation 3× (-run TestIdleShutdown_TouchResets -count=3) — PASS/PASS/PASS. Re-ran the full pkg/lifecycle package 3× more (-count=1 each) — PASS/PASS/PASS. **6/6 re-runs PASSED; only the original full-module sweep run FAILED once.**

This is captured empirical evidence of a load/timing-sensitive test (the FAIL occurred only when go test ./... was running many containers packages concurrently, i.e. under host CPU/scheduling contention). Root cause is **UNCONFIRMED** per §11.4.6 — no source was touched to investigate further (out of this stream’s read-only scope). This is reported as a real finding, not smoothed over: it is a genuine, if non-deterministic, test-suite result the conductor should track (candidate root causes to investigate: a wall-clock/timer-based idle-reset assertion that is not robust to scheduling delay under concurrent test load).

Full detail + reproducibility-check transcript:
submodules_containers.log.

2. submodules/llm_orchestrator — gofmt reports 6 files needing formatting

```
pkg/agent/agent.go
pkg/agent/agent_test.go
pkg/agent/simple_pool.go
pkg/parser/parser.go
pkg/parser/parser_security_test.go
pkg/protocol/message.go
```

Not reformatted (read-only mandate). Real finding — gofmt -l returned non-empty, meaning the recent formatting sweep referenced in the task prompt did not (or no longer) cover this module.

3. submodules/vision_engine — gofmt reports 16 files needing formatting

```
cmd/visiondescribe/main.go
pkg/analyzer/i18n_defaults.go
pkg/analyzer/types.go
pkg/analyzer/types_test.go
pkg/config/i18n_callsites_test.go
pkg/graph/graph.go
pkg/graph/graph_automation_test.go
pkg/llmvision/anthropic.go
pkg/llmvision/astica.go
pkg/llmvision/i18n_defaults.go
pkg/llmvision/provider.go
pkg/llmvision/provider_test.go
pkg/opencv/i18n_defaults.go
pkg/opencv/orb_vision_test.go
pkg/remote/deployer_test.go
pkg/remote/remote.go
pkg/remote/remote_test.go
```

Not reformatted (read-only mandate). Real finding, same class as #2.

Coverage-gap check (§11.4.169)

No swept module is entirely without tests — every module produced at least one ok package result. Sub-package [no test files] results are present in several modules (e.g. cmd/ entryptpoint packages, interface-only packages, fixture directories) — these are normal, not whole-module gaps. No module-level “no test files” coverage gap was found in this sweep.

Evidence files

- docs/qa/20260710-owned-bricks-test-sweep/COVERAGE_SWEEP.md (this file)
- docs/qa/20260710-owned-bricks-test-sweep/submodules_challenges.log
- docs/qa/20260710-owned-bricks-test-sweep/submodules_containers.log
- docs/qa/20260710-owned-bricks-test-sweep/submodules_doc_processor.log

- docs/qa/20260710-owned-bricks-test-sweep/submodules_llm_orchestrator.log
- docs/qa/20260710-owned-bricks-test-sweep/submodules_llm_provider.log
- docs/qa/20260710-owned-bricks-test-sweep/submodules_llms_verifier.log
- docs/qa/20260710-owned-bricks-test-sweep/submodules_llms_verifier_llm-verifier.log
- docs/qa/20260710-owned-bricks-test-sweep/submodules_ota-artifact-validator.log
- docs/qa/20260710-owned-bricks-test-sweep/submodules_ota-protocol.log
- docs/qa/20260710-owned-bricks-test-sweep/submodules_ota-rollout-engine.log
- docs/qa/20260710-owned-bricks-test-sweep/submodules_ota-telemetry-schema.log
- docs/qa/20260710-owned-bricks-test-sweep/submodules_security.log
- docs/qa/20260710-owned-bricks-test-sweep/submodules_vision_engine.log

No HTML/PDF siblings exported (conductor's responsibility per task scope).

Conductor reconciliation (2026-07-10, independent spot-check §11.4.2/§11.4.6)

Re-ran the two flagged findings myself against the live tree before accepting this sweep:

- **Finding 1 — containers TestIdleShutdown_TouchResets:** re-ran in isolation → ok pkg/lifecycle 0.306s (PASS). Combined with the sweep's 6/6 isolated reruns, this is a **timing-sensitive test that failed once only under the parallel-sweep host contention** (all 13 modules testing at once → CPU/scheduler pressure delayed an idle-timer). Root cause **UNCONFIRMED** (§11.4.6 — not asserting a cause without forensic proof); classification PENDING_FORENSICS. **NOT release-blocking** for the OTA control plane — it is a lifecycle-timer test in the generic vasic-digital/containers brick, not a control-plane defect, and does not reproduce without induced load. Tracked follow-up: harden the test's timer margin or gate it on a

load-detector; owner = a future operator-batchable submodule window (same window as the mirror-fork gofmt below).

- **Findings 2+3 — gofmt on llm_orchestrator (6 files) + vision_engine (17 files; sweep said 16, live count 17):** confirmed real. These are **exactly the two bricks already DEFERRED** from the wave-4 gofmt canonicalization sweep for a pre-existing multi-mirror fork (see CONTINUATION §1 / docs/research/submodule_gofmt_vet_sweep_20260710/FINDINGS.md Rev 2). Their gofmt is intentionally pending the careful §11.4.113 merge-onto-latest-mirror convergence (no force-push), which is operator-batchable — NOT a new regression. Consistent with known state.

Net: all 13 owned Go bricks build + vet clean; 11/13 fully gofmt-clean + tests-PASS; the 2 gofmt-pending bricks are the known deferred-mirror pair; one timing-flaky containers test is PENDING_FORENSICS, non-blocking. No new release blocker — the loop continues (§11.4.4 not triggered: no reproduced product defect).